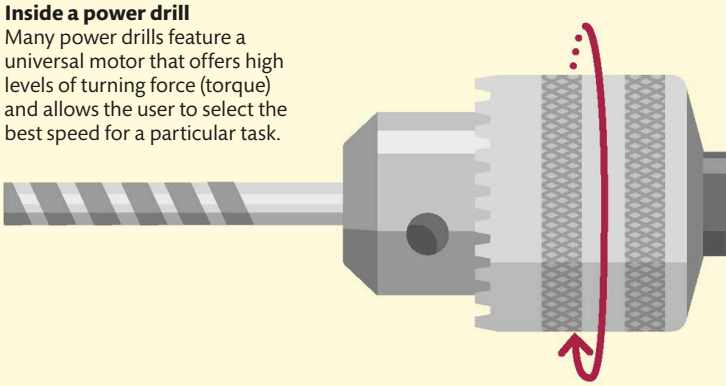
Universal motors

In a universal electric motor, the permanent magnet is replaced by an electromagnet made of a number of windings of wire through which current can flow. This produces a magnetic field, inside which a coil, called an armature, rotates. Both the armature and the wire windings surrounding it receive the same electric current because they are wired in series. This means a universal motor can run on both DC and AC electricity.

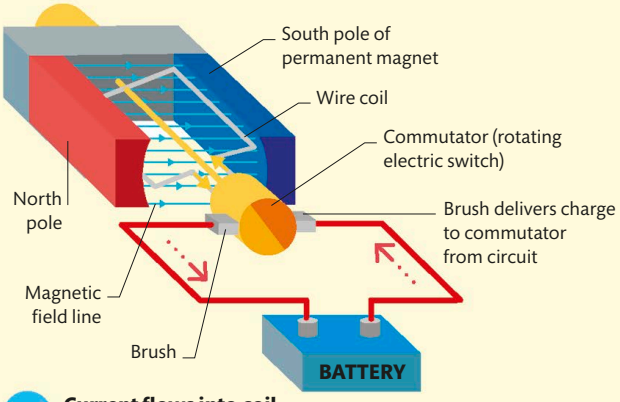
Inside a power drill
Many power drills feature a universal motor that offers high levels of turning force (torque) and allows the user to select the best speed for a particular task.



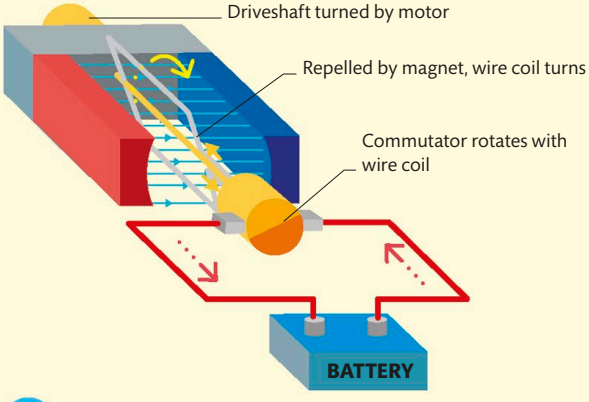
Motors

Electric motors use the forces of attraction and repulsion between an electric current and a magnetic field to create turning movement. Motors vary in size, from microscopic actuators inside electronic gadgets to giant power plants that propel large ships.

AROUND 45 PERCENT OF ALL ELECTRICITY CONSUMED POWERS ELECTRIC MOTORS



1 Current flows into coil
An electric current flows into a wire coil positioned between the poles of a permanent magnet. The coil becomes an electromagnet.



2 Wire coil turns
Repelled by the magnet's like poles, the coil turns away. After a quarter turn, the unlike poles attract, forcing the coil into a half-turn.

How an electric motor works

In many motors, a coil of wire moves through the magnetic field produced by a stationary magnet. When current flows through the coil, it becomes an electromagnet with north and south poles. The coil swings around to align its poles with those of the permanent magnet. A commutator reverses the coil's current every half-turn to switch the coil's poles and keep it spinning in the same direction. The coil is connected to a driveshaft, which transmits the motor's turning force to components, such as wheels.

HOW FAST DO DC MOTORS SPIN?

An average DC motor spins at 25,000 rpm, but some motors, such as those in vacuum cleaners, can reach up to 125,000 rpm.